

Curriculum Vitae: Rodrigo Avalos

Personal information

Name: Rodrigo Avalos.

Date of Birth: September 22, 1989.

Place of Birth: Villa Gesell, Buenos Aires, Argentina.

Nationalities: Argentinian and Italian.

E-mail(s): rodrigo.avalos@uni-rostock.de and rdravalos@gmail.com

Academic positions

Unbefristeter wissenschaftlicher Mitarbeiter (Permanent Academic Staff) at the Department of Mathematics of the University of Rostock, Germany. Since April 2026

Wissenschaftlicher Mitarbeiter (Postdoc) at the Department of Mathematics of the University of Tübingen, Germany. October 2024 - March 2026

Alexander von Humboldt Postdoctoral Fellow at the Department of Mathematics of the University of Tübingen, Germany. October 2023 - September 2024

Alexander von Humboldt Postdoctoral Fellow at the Institute of Mathematics of the University of Potsdam, Germany. October 2022 - September 2023

Postdoctoral FUNCAP position at the Department of Mathematics of the Federal University of Ceará, Brazil. September 2021 - September 2022

Postdoctoral CAPES/PNPD position at the Department of Mathematics of the Federal University of Ceará, Brazil. October 2017 - March 2021

Education

Habilitation July 2023 - July 2026

Department of Mathematics of the University of Tübingen (Germany) - Mentor: Prof. Dr. Carla Cederbaum.

Doctorate March 2014 - October 2017

PhD in Physics (Gravitation and Cosmology) awarded by the Federal University of Paraíba (UFPB), Brazil.

Degree March 2008 - March 2014

Licenciado en Ciencias Físicas, FCEyN - Universidad Nacional de Mar del Plata (equivalent to an integrated master's degree in physics, awarded by the National University of Mar del Plata -UNMDP-, Argentina).

Areas of research

My main research interests lie in the overlap between differential geometry, geometric analysis and physics, focusing on geometric problems arising in general relativity. I am specially interested in the analysis of initial data sets in general relativity and geometric partial differential equations (PDEs) related to these topics. These include Yamabe-type equations and other PDEs arising in conformal geometry, such as Q -curvature conformal deformations, as well as low regularity versions of these problems. I am also interested in the understanding of asymptotic invariants arising in general relativity and their relation to these geometric PDE problems.

Grants, scholarships and fellowships

- Post-doctoral fellowship awarded by Alexander von Humboldt Foundation (10/2022 - 09/2024).
- Post-doctoral fellowship awarded by the Fundação Cearense de Apoio ao Desenvolvimento Científico e Tecnológico (FUNCAP) - Research position at the Department of Mathematics of the Federal University of Ceará (09/2021 - 09/2022).
- Post-doctoral fellowship awarded within the Brazilian Postdoctoral Program (PNPD) financed by Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) - Research position at the Department of Mathematics of the Federal University of Ceará (10/2017 - 08/2021).
- Doctoral fellowship jointly awarded by Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) and Centro Latino-Americano de Física (CLAF) (From 2014-2017).
- “Beca de estímulo a la vocaciones científicas”, granted by the Consejo Interuniversitario Nacional (One-year undergraduate scholarship granted by the Argentinian agency CIN - from 2012 to 2013).
- “Beca de estímulo a la vocaciones científicas”, granted by the Consejo Interuniversitario Nacional (One-year undergraduate scholarship granted by the Argentinian agency CIN - from 2011 to 2012).

Awards

- “Graduado Sobresaliente de la UNMP” (Award given by the National University of Mar del Plata to distinguished students who finish their graduation with an exceptional performance: average 9,5/10).

Language skills

- Spanish - Mother language.
- English - C1 level.
- Portuguese - C1 level.
- German - B2 level.

Participation in examination boards

- First evaluator (advisor) for the Master’s thesis “Asymptotic Analysis of Isolated Gravitational Systems and the ADM Center of Mass”. Author: Rafael Olapraph. Institution:

University of Tübingen (2025).

- Second evaluator for the Master’s thesis “Approaches to prove Hamilton’s conjecture on Ricci-pinched three-manifolds”. Author: Ariadna León Quiros; Advisor: Prof. Dr. Gerhard Huisken. Institution: University of Tübingen (2024).
- Second evaluator for the Bachelor’s thesis “On Capacity and ADM Mass in Rough Asymptotically Flat Initial Data Sets”. Author: Robert Lenschow; Advisor: Prof. Dr. Jan Metzger. Institution: University of Potsdam (2023).
- Participation in the evaluation committee of the PhD thesis: “Graph solitons of the mean curvature flow”, Author: Renivaldo Sodré de Sena. Advisor: Prof Dr. Luciano Mari. Institution: Federal University of Ceará (2018).

Supervisions

Concluded

- Rafael Olapraph, “*Asymptotic Analysis of Isolated Gravitational Systems and the ADM Center of Mass*”, Master’s in Mathematical Physics, University of Tübingen, concluded in October 2025.

Ongoing

- Noah Migoski, “*Point particle motion and the geodesic postulate in general relativity*”, Master’s in Mathematical Physics, University of Tübingen, expected conclusion by September 2026.

List of Publications

Preprints under review

- R. Avalos, A. Cogo and A. Royo Abrego, *Regularity of conformal structures on closed 3-manifolds*, arXiv:2511.00178 (2025).
- R. Avalos, A. Cogo and A. Royo Abrego, *Conformal Green Functions and Yamabe Metrics of Sobolev Regularity*, arXiv:2507.01674 (2025).
- R. Avalos, J. H. Lira and N. Marque, *Einstein Type Systems on Complete Manifolds*, arXiv:2201.08347v2 (2022, revised 2025).¹

Published papers

- R. Avalos, *Sobolev regularity of compactified 3-manifolds and the ADM Center of Mass*, accepted for publication in *Annals of PDE* (2026). ArXiv:2403.04034. <https://doi.org/10.1007/s40818-026-00252-4>
- R. Avalos, P. Laurain and N. Marque, *Rigidity Theorems for Asymptotically Euclidean Q -singular Spaces*, Adv. Math. 490, 110850 (2026). <https://doi.org/10.1016/j.aim.2026.110850>
- R. Avalos, E. Ling and A. Piubello, *On energy and its positivity in spacetimes with an expanding flat de Sitter background*, Class. Quantum Grav. 43, 095006 (2026). <https://doi.org/10.1088/1361-6382/ae5eb7>

¹This received a positive first review in *Letters Mathematical Physics* and is now undergoing review of its revised version.

- R. Avalos, J. H. Lira and N. Marque, *Energy in Fourth Order Gravity*, Ann. Henri Poincaré 26, 597–673 (2025). <https://doi.org/10.1007/s00023-024-01440-3>
- R. Avalos, *On the Rigidity of Cosmological Space-times*, Lett Math Phys 113, 98 (2023). <https://doi.org/10.1007/s11005-023-01720-9>
- R. Avalos, P. Laurain and J. H. Lira, *A Positive Energy Theorem for Fourth-Order Gravity*, Calc. Var. 61, 48 (2022). <https://doi.org/10.1007/s00526-021-02152-w>
- R. Avalos and J. H. Lira, *Einstein-type elliptic systems*, Ann. Henri Poincaré 23, 3221–3264 (2022). <https://doi.org/10.1007/s00023-022-01180-2>
- R. Avalos, *Positive Energy Theorems in Fourth-Order Gravity*, In: Albuje, A.L., Caballero, M., García-Parrado, A., Herrera, J., Rubio, R. (eds) Developments in Lorentzian Geometry. GELOMA 2021. Springer Proceedings in Mathematics & Statistics, vol 389. Springer, Cham (2022). https://doi.org/10.1007/978-3-031-05379-5_3
- R. Avalos and A. Freitas, *The Pohozaev-Schoen Identity on Asymptotically Euclidean Manifolds: Conservation Identities and their applications*, Annales de l’Institut Henri Poincaré C, Analyse non linéaire, 38, 6, 1703-1724 (2021). <https://doi.org/10.1016/J.ANIHPC.2021.01.002>
- R. Avalos, *The Thin-Sandwich Problem in General Relativity*, In: Ortegon Gallego F., García García J. (eds) Recent Advances in Pure and Applied Mathematics. RSME Springer Series, vol 4. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-41321-7_6
- R. Avalos and J. H. Lira, *Reduced thin-sandwich equations on manifolds euclidean at infinity and on closed manifolds: existence and multiplicity*, J. Math. Phys., 61, 122501 (2020). <https://doi.org/10.1063/5.0008749>
- R. Avalos, I. P. Lobo, T. Sanomiya, C. Romero, *On the Cauchy Problem for Weyl-Geometric Scalar-Tensor Theories of Gravity*, J. Math. Phys. 59, 062502 (2018). <https://doi.org/10.1063/1.5017848>
- R. Avalos, F. Dahia, C. Romero, *On the C^k -embedding of Lorentzian manifolds in Ricci-flat spaces*, J. Math. Phys. 59, 052503 (2018). <https://doi.org/10.1063/1.5017492>
- R. Avalos, F. Dahia, C. Romero, *A Note on the Problem of Proper Time in Weyl Space-Time*, Found. Phys., 48, 253-270 (2018). <https://doi.org/10.1007/s10701-017-0134-z>
- R. Avalos, F. Dahia, C. Romero, J. H. Lira, *On the proof of the Thin Sandwich Conjecture in arbitrary dimensions*, J. Math. Phys. 58, 102502 (2017). <https://doi.org/10.1063/1.4989573>
- R. Avalos, F. Dahia, C. Romero, *On the embedding of Weyl manifolds*, J. Math. Phys. 58, 012502 (2017). <https://doi.org/10.1063/1.4973751>

Other publications

- R. Avalos, *Regularity of compactified 3-manifolds and the ADM Center of Mass*, Oberwolfach Report No. 36/2024. Workshop: Mathematical Aspects of General Relativity

Work in preparation

- R. Avalos, D. Maxwell, S. Senthil-Velu and O. Vicanek Martinez *On the extrinsic geometry of isolated systems at spatial infinity*, (expected conclusion: September 2026).

- R. Avalos, A. Cogo and A. Royo Abrego, *Bach-flat metrics and regularity theory of conformal structures*, (expected conclusion: September 2026).
- R. Avalos and C. Cederbaum, *On the (non)-existence of Schwarzschildian coordinates for boosted initial data sets*, (expected conclusion: October 2026)
- R. Avalos and N. Marques, *A note on the L^2 -regularity of solutions to the conformally formulated coupled Einstein constraint equations*, (expected conclusion: December 2026)

Teaching experience

- Tutor for “Analysis II” - summer semester 2026. Bachelor’s course at the University of Rostock, Germany. Workload of 2 hrs/week.
- Tutor for “Mathematics 2 for Engineers” - summer semester 2026. Bachelor’s course at the University of Rostock, Germany. Workload of 6 hrs/week.
- Instructor for “Introduction to Partial Differential Equations” - winter semester 2025/26. Master’s course at the University of Tübingen, Germany. Workload of 6 hrs/week.
- Instructor for “Riemannian Geometry” - summer semester 2025. Master’s course at the University of Tübingen, Germany. Workload of 4 hrs/week.
- Instructor for “Introduction to Partial Differential Equations” - winter semester 2024/25. Master’s course at the University of Tübingen, Germany. Workload of 6 hrs/week.
- Instructor for “Mathematical Relativity” - summer semester 2024. Master’s course at the University of Tübingen, Germany. Workload of 6 hrs/week.
- Instructor for the mini course “The Einstein constraint equations” during the XI International Meeting on Lorentzian Geometry, held at at Universidad Autónoma de Yucatán (in Mérida, México) from January 29th to February 2nd, 2024.
- Instructor for the course “The Einstein constraint equations” - summer semester 2023. Graduate level course at the University of Potsdam, also offered as a hybrid course within the Master’s program in Mathematical Physics at the University of Tübingen, Germany. Workload of 4 hrs/week.
- Instructor for the advanced mini-course “The Einstein Constraint Equations” during the 33rd-Brazilian Colloquium of Mathematics, Organized by the “Instituto de Matemática Pura e Aplicada (IMPA)”, Rio de Janeiro, held from August 2 - 6, 2021.
- Instructor for the course “Lorentz Geometry with Applications to General Relativity” - first semester 2017. Graduate level course at the Federal University of Paraíba (UFPB), Brazil. Workload 2 hrs/week.
- Teaching assistant for “Physics I” (classical mechanics) - both semesters in 2015. Undergraduate course at the Federal University of Paraíba (UFPB), Brazil. Workload 2 hrs/week.
- Teaching position (Docente interino) for “Physics III” (wave mechanics) - both semesters of 2013. Undergraduate exercise classes at the National University of Mar del Plata, Argentina. Workload 4 hrs/week.
- Undergraduate teaching assistant (Auxiliar adscripto) for “Physics I” - second semester 2010. Undergraduate exercise classes at the National University of Mar del Plata, Argentina. Workload 4 hrs/week.
- Undergraduate teaching assistant (Auxiliar adscripto) for “Physics I” - second semester 2009. Undergraduate exercise classes at the National University of Mar del Plata, Argentina. Workload 4 hrs/week.

Lecture notes

- Lecture Notes on Partial Differential Equations available via ²
- Lecture Notes on Mathematical Relativity available via ³
- Lecture Notes on Riemannian Geometry available via ⁴
- R. Avalos and J. H. Lira, *The Einstein Constraint Equations*, Editora do IMPA, Rio de Janeiro, Brazil (2021).⁵

Selected talks

- *Yamabe metrics of Sobolev regularity*, Annual 2025 ÖMG-DMV Meeting, held at the Johannes Kepler University, Linz, Austria, 2025 (Lecture).
- *Asymptotic analysis of isolated gravitational systems*, Conference “Geometry at Infinity”, Max Planck Institute for Mathematics in the Sciences, Leipzig, Germany, 2025 (Lecture).
- *Asymptotic analysis of isolated gravitational systems*, within the section “Chronology and causality, CTCs” of the conference “Time Machine Factory”, held in Turin, Italy 2024 (Lecture).
- *Regularity of compactified 3-manifolds and the ADM Center of Mass*, Workshop “Mathematical Aspects of General Relativity”, held at Mathematisches Forschungsinstitut Oberwolfach, Germany, 2024 (Lecture).
- *A Q-curvature positive energy theorem with applications to rigidity phenomena*, Workshop: “Mass, the Einstein Constraint Equations, and the Penrose Inequality Conjecture”, Simons Center for Geometry and Physics, Stony Brook, United States of America, 2023 (Lecture).
- *The Einstein Constraint Equations of General Relativity*, Department of Mathematics of the University of Tübingen, Germany, 2023 (Kolloquium talk).
- *A Q-curvature positive energy theorem*, 13th Central European Relativity Seminar, held in Stockholm, Sweden 2023 (short talk).
- *The coupled Einstein constraint equations on non-compact manifolds*, interdisciplinary junior scientist workshop: Mathematical General Relativity, Wildberg, Germany (2023) (short talk).
- *A Q-curvature positive energy theorem and rigidity of Q-singular asymptotically Euclidean manifolds*, Workshop on Differential Geometry, held at the University of Granada, Spain, 2023 (Lecture).
- *Positive Energy Theorems in Fourth-Order Gravity*, X International Meeting on Lorentzian Geometry, GeLoCor 2021, held in Córdoba, Spain, 2021 (Lecture).
- *Advances on the Einstein constraint equations*, Cycle of Talks on Geometry, held at the Federal University of Paraíba, Brazil, 2019 (Lecture).

²<https://www.dropbox.com/scl/fi/q6vdx89outh7nl6e1c61s/Lecture-Notes-on-PDEs-Rodrigo-Avalos.pdf?rlkey=gi600y2opicyhxgz138us29im&st=5i367q84&dl=0>

³<https://www.dropbox.com/scl/fi/ut0bhv4pqmm1m44altepa/Lecture-Notes-on-Mathematical-Relativity-Rodrigo-Avalos.pdf?rlkey=y0gxewr54ogt14j5ipxzxpybw&st=knubcz4z&dl=0>

⁴<https://www.dropbox.com/scl/fi/ybl2u8fhvl40lvhfmmqj7/Lecture-Notes-Riemannian-Geometry.pdf?rlkey=1f0ouckdtfe0q980ygyb46exg&st=xbeyk6vh&dl=0>

⁵Updated and corrected version can be requested via e-mail.

- *Advances on the Einstein constraint equations*, 32 Brazilian Colloquium of Mathematics held at the Instituto Brasileiro de Matemática Pura e Aplicada (IMPA) in Rio de Janeiro, Brazil, 2019 (Lecture).
- *An analysis of the reduced thin-sandwich equations of general relativity*, Three days workshop in mathematical general relativity, held at the Institut Camille Jordan in Lyon, France, 2019 (Lecture).
- *An analysis of the reduced thin-sandwich equations of general relativity*, II Joint Meeting Spain-Brazil in Mathematics held in Cádiz, Spain: Special session on Lorentzian geometry and its applications, 2018 (Lecture).
- *On the proof of the thin sandwich conjecture in arbitrary dimensions*, X Workshop on Geometric Analysis, held at the Federal University of Ceará, Brazil 2017 (Lecture).

Administrative duties

- Joint organiser of the “43. Süddeutsches Kolloquium über Differentialgeometrie”, held in Tübingen on May 7-8, 2026. <https://sueddeutsches-kolloquium.squarespace.com>
- Joint organiser of the seminar on “Geometrische Analysis, Differentialgeometrie und Relativitätstheorie” at the Mathematics Department of the University of Tübingen during the winter semester 2025/26, summer semester 2025, winter semester 2024/2025, summer semester of 2023 and winter semester 2022/2023.
- Joint organiser of the 14th Central European Relativity Seminar, held in Tübingen on February 14-16, 2024. <https://cers.univie.ac.at/cers14/index.html>
- Student member in the Department Council of the Physics Department at the National University of Mar del Plata during 2013.

Upcoming events

- I will be organising the workshop “PDEs and Conformal Geometry” together with Prof. Volker Branding at the University of Rostock during September 29 - October 01, 2027. <https://www.mathematik.uni-rostock.de/pde-conformal-geometry27/>

Research projects

As a research associate/postdoctoral fellow:

- **Project:**
Geometrically defined asymptotic coordinates in general relativity 2022 - 2025
Role: Research team member from 2023-2025.
Project manager(s): Carla Cederbaum (University of Tübingen) and Jan Metzger (University of Potsdam).
Sponsor(s): Deutsche Forschungsgemeinschaft (DFG) - Germany.
- **Project:** Relativistic Initial Data Sets: Analysis of Conserved Quantities and Initial Data Constructions 2022 - 2024
Role: Project manager.
Sponsor(s): Alexander von Humboldt Foundation - Germany.

- **Project:** Nonlinear PDEs and Geometry of Interfaces 2020 - 2022
Role: Research team member from 2020-2022. Within the project, which also included interactions with French universities, I was in charge of the regular interactions with Prof. Paul Laurain's group at the University of Paris, including developing the research strategies and setting up research group meetings.
Project manager(s): Marcos Petrúcio de A. Cavalcante (Federal University of Alagoas - UFAL).
Sponsor(s): Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) - Brazil.
- **Project:** Geometric Analysis and Singularity Theory on Stratified Spaces 2018 - 2021
Role: Research team member from 2018-2021. Within the project, I was in charge of the regular interactions with Prof. Paul Laurain's group at the University of Paris, including developing the research strategies and setting up research group meetings.
Project manager(s): Jorge Herbert Soares de Lira (Federal University of Ceará - UFC) and Paul Laurain (University of Paris).
Sponsor(s): Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) and Comité Français d'Évaluation de la Coopération Universitaire et Scientifique avec le Brésil (COFECUB) - Brazil and France.
- **Project:** Programa de Apoio a Núcleos de Excelência 2017 - 2018
PRONEX/FUNCAP/CNPq Núcleo de análise geométrica, singularidades e combinatória (Research program on geometric analysis, singularities and combinatorics).
Role: Research team member during 2018.
Project manager(s): Gregório Pacelli Feitosa Bessa (Federal University of Ceará - UFC).
Sponsor(s): Fundação Cearense de Apoio ao Desenvolvimento Científico e Tecnológico (FUNCAP) and Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) - Brazil.

As a PhD student:

- **Project:** On some geometric aspects of gravity: Space-time embeddings, Wheeler's thin-sandwich conjecture and Weyl manifolds 2014-2017
Role: Project co-manager together with my thesis advisor Prof. Dr. Carlos A. Romero Filho (Federal University of Paraíba, Brazil).
Sponsor(s): Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) and Centro Latino-Americano de Física (CLAF) - Brazil.

Editorial activities

- Reviewer for Communications in Mathematical Physics, Advances in Theoretical and Mathematical Physics, Communications in Analysis and Geometry, Mathematische Zeitschrift, Mathematical Reviews and zbMATH.
- Reviewer for the Banff International Research Station.